LESSONS LEARNED — P2B CONSOLIDATION

R3 College · 3R31 / 3R32 / 3R33 · project-closeout-oriented record · 25 October 2026

1. A taxonomy does not determine the number of implemented agents

Faculty asked why four rather than three, five or X. We separated A1–A4 categories from implementation instances, identified four functional responsibilities and compared configurations qualitatively. Four agents remain a revisable initial design, not an empirically optimal number.

2. Real–digital convergence must be operationally defined

Progressive fidelity, synchronization and integration make the aspiration of one coordinated sociotechnical system testable. Real and digital retain distinct F/S/P/C descriptions and human decision authority.

3. A tetrahedron is more than four independent lists

We confronted six pairwise relationships, four three-dimensional faces and a joint synthesis. This revealed cross-process effects and governance limitations missed by isolated descriptions.

4. Feasibility must change architectural choices

A real third iterative level needs a visible outcome: adopt, condition, modify or reject each alternative, with evidence-based rationale. Real-time synchronization for every variable and unrestricted physical autonomy were not adopted.

5. Bibliometric counts require lineage and boundaries

The 2025 F/S/P/C lexical queries and the 2026 technology-focused queries were kept separate. The 3R33 documentary reuse is not counted as an independent search. Retrieval counts do not establish scientific gaps by themselves.

6. A design decision is not an experimentally validated result

We distinguished documented source content, AI-assisted recommendations, human approvals, and future validation. This avoids implying existing production deployment or verified economic and causal benefits.

7. Track decisions as presentation evidence

The human-approved P2B-01–11 decisions describe WHAT changed, WHICH alternatives were considered, WHY they were selected, the groups’ contributions and the role of GPT-6 (Astra). Approval by all twelve members has not been asserted.

Outstanding items / Pendências

Remaining controls: exact bibliographic checking; line-by-line PT–EN equivalence; stakeholder and field validation; economic feasibility; project closeout and final group review.